

Modeling and Analysis of Bipedal Wheel-legged Robot with Sprawling Legs Mechanism

Zulkifli Mansor, Addie Irawan*, Nor Maniha Abdul Ghani and Azrul Azim Abdullah Hashim
Robotics, Intelligent Systems & Control Engineering (RiSC) research group
Faculty of Electrical & Electronics Engineering Technology,
Universiti Malaysia Pahang Al-Sultan Abdullah,
 26600, Pekan, Pahang, Malaysia
 addieirawan@ump.edu.my*

Abstract— This paper presents the modeling and analysis of the novel Bipedal Wheel-Legged Robot with Sprawling Legs Mechanism (BWLS) utilizing multibody dynamics approaches and the proposed Coxa-Based Coordinate System (CCS) in kinematics modeling. The primary focus lies in transitioning the conceptual model of the BWLS into the Simscape Multibody (SM) environment. The proposed CCS is subjected to verification through the implementation of a Hierarchical Control Framework (HCF) utilizing a simple stand-up input trajectory. Through the analysis, initial angular positions and stable range angular joints are determined, revising previous determinations based on hardware structure. Furthermore, employing the Forward Kinematics (FK) model, the stable boundary is identified through spatial leg motion plots. The reliability of the proposed CCS kinematics model is affirmed, rendering the model ready for further in-depth studies and applications.

Keywords— *wheel-legged robot, bipedal robot, kinematics, multibody dynamics.*

I. INTRODUCTION

As the need intensifies for ground mobile assistive systems to navigate complex landscapes such as islands and mountainous regions, bipedal walking robots have risen to prominence in the realm of research and development. These mechanical marvels boast dynamic stability, superior energy efficiency, and a gait reminiscent of human locomotion, rendering them ideal for deployment across diverse indoor and outdoor scenarios. Their adeptness at traversing uneven terrains and confined spaces earmarks them as indispensable allies in exploration and human aid endeavors. Yet, the pursuit of perfection in their design faces ongoing scrutiny, with persistent inquiries delving into matters of resilience, self-sufficiency in power, and the mitigation of mechanical wear [1, 2]. These robots imitate the bipedal locomotion characteristic of humans and other mammals, striving to replicate the natural rhythm of movement[3]. To achieve this, intricate control algorithms, sensory feedback mechanisms, and mechanical configurations are essential to ensure stability and flexibility [2]. Introducing a hybrid approach, wherein wheels are integrated at the extremities of the bipedal legs, enhances the robot's capabilities by amalgamating the agility of legged locomotion with the speed and efficiency of wheeled systems, along with the adaptability inherent in legged robots. However, motion planning for Bipedal Wheel-legged Robots (BWLR), which combine wheeled and legged locomotion systems, presents distinctive challenges exceeding those of traditional high-dimensional and

nonlinear platforms. This hybridization offers enhanced flexibility in navigating varied terrains but introduces complexities due to underactuation and coordinating fewer control inputs with degrees of freedom (DoF) [4].

In recent years, there have been many advancements in the hardware platforms for wheeled bipedal robots, with several developed models presented in literature such as reported in[5-7]. However, there remains room for further research and investigation into the planning and control strategies needed for these robots to perform more complex motions. Numerous studies have focused on enhancing the stability and agility of robots through refined design and control mechanisms [8-12]. Others have delved into exploring compliant leg mechanisms in legged robots, analyzing workspace fatigue, and leveraging compliant joints for locomotion[13, 14]. Concurrently, research has expanded to encompass all-terrain wheel-legged robots, aiming to improve adaptability in complex environments [15-18], mirroring trends seen in other legged robot research [19-22].

The development of hardware capabilities has outpaced the control software and algorithms required to fully utilize these hybrid robots. More work must be done on enabling intelligent planning and movement in wheeled biped robots with diverse capabilities. The software-in-loop concept in control study with the integration between the solid model and the numerical software to presenting the 3-dimensional (3D) simulation is the one of the approaches that can pre-deployment of advance planning and control strategy of the complex mechanism robot such as BWLR.

Therefore, study takes the initiative to design and model a novel bipedal wheel-legged robot with a sprawling leg (BWLS) mechanism, as patented in[23]. Surprisingly, despite extensive research in robotics, this specific mechanism remains relatively unexplored. Most existing approaches focus on integrating wheels and legs onto a single platform, aiming to combine the advantages of both modes of locomotion, including energy efficiency [24, 25]. The paper's primary contribution lies in establishing a transition solid model from the conceptual model of BWLS in computer aid drawing (CAD) form to the SIMULINK MATLAB® Simscape format using Simmechanics libraries. Additionally, the Coxa-coordination concept is applied to kinematics modeling.

The remainder of this paper is organized as follows: Section II presents an overview of the proposed BWLS design model and its configuration. Then, Section III discusses the proposed Coxa-based coordination kinematics with leg motion trajectory input. Section IV analyzes the

stable range of angular input for each joint of the robot's motion and examines the impact of integrating the kinematic model with the generated Simscape model. Here, the robot's locomotion performance through its Cartesian leg motion is discussed, and kinematics are verified between both inverse and forward models. Finally, the paper is concluded in Section IV.

II. OVERVIEW ON BWLS SYSTEM

The bipedal wheel-legged robot with sprawling leg mechanism (BWLS) is meticulously designed to emulate human-like walking capabilities while maintaining the agility necessary for skiing maneuvers. This innovative robot incorporates a flexible sprawling joint mechanism with Coxa mechanisms at the hip area of each leg. With four degrees of freedom (DoF) per leg, including wheels that function as analogs to ankles during movement, BWLS achieves unparalleled versatility. Its sprawling leg design allows seamless transitions between sliding, humanoid leg motions, and anthropomorphic postures[23]. To power these complex movements, BWLS is equipped with high-torque and high-speed direct current (DC) motors on each joint. Detailed specifications regarding leg dimensions and joint ranges provide a comprehensive overview of the robot's mechanical prowess.

TABLE I. BWLS LEG DIMENSION AND JOINT RANGE ESTIMATION

Joint	Link	Length (m)	Range of angle (Deg) (Estimation)
Coxa-Hip	l_1	0.00	$-120^\circ \leq q_1 \leq 60^\circ$ (Leg 1) $-60^\circ \leq q_1 \leq 120^\circ$ (Leg 2)
Hip-Knee	l_2	0.292	$50.4^\circ \leq q_2 \leq 135^\circ$
Knee-Wheel	l_3	0.375	$295^\circ \leq q_3 \leq 315^\circ$
Wheel	l_4/R	0.05	$145^\circ \leq q_4 \leq 60^\circ$ (walk mode)

For modeling purposes, the Simscape Multibody (SM) within MATLAB® serves as the foundation, meticulously prepared for integration with Quadratic Programming (QP) developed in SIMULINK. The objective is to streamline the model complexity, illustrated in Figure 1, by transitioning the robot into a conceptual model depicted in Figure 2(a). This remodeling endeavor aims to mitigate the extensive matting modules inherent in the SM version, enhancing computational efficiency. Figure 2(b) showcases the refined SM version of the BWLS robot model, optimized for improved performance and ease of analysis.

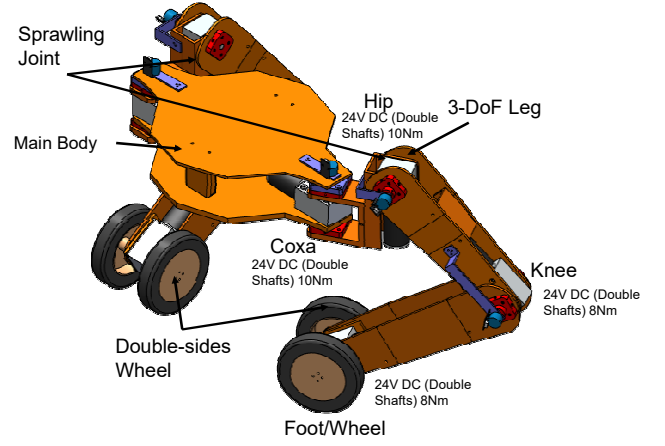


Fig. 1. Overall control system diagram of TPG configuration and setup for a single finger control

In the comprehensive hierarchical control framework (HCF) governing the BWLS model, the establishment of a robust kinematics model stands as a fundamental prerequisite. Figure 3 delineates the foundational structure of the HCF for BWLS, featuring a trajectory module facilitating maneuvering and motion planning, a kinematics model facilitating translation between Cartesian and angular motion inputs/outputs, and a position control system ensuring precise joint input accuracy. This intricate framework underscores the meticulous attention to detail in engineering a system capable of seamless coordination and precise control over the robot's dynamic behavior.

III. COXA-BASED COORDINATE SYSTEMS

A. Initial based on Coxa of the Leg

For the coxa-based coordination system (CCS), the coordinate system for each leg is designed using the coxa joint as the basis for calculating kinematic models. The initial coordinate for each individual leg is calculated according to the geometry of the leg, centered on the initial coxa angle (θ_c), as follows;

$$x_{0_n} = (\delta_{n,m} - \delta_{n,m})l_c \sin(\theta_c) \quad (1)$$

$$y_{0_n} = (\delta_{n,m} - \delta_{n,m})l_c \cos(\theta_c) \quad (2)$$

with;

$$\delta_{n,m} = \begin{cases} 1 & \text{if } n = m \\ 0 & \text{else} \end{cases}$$

where $n = m = 1, 2$ is the number of leg and $l_c = 0.25m$ is the length between Coxas as can be depicted in Fig. 2.

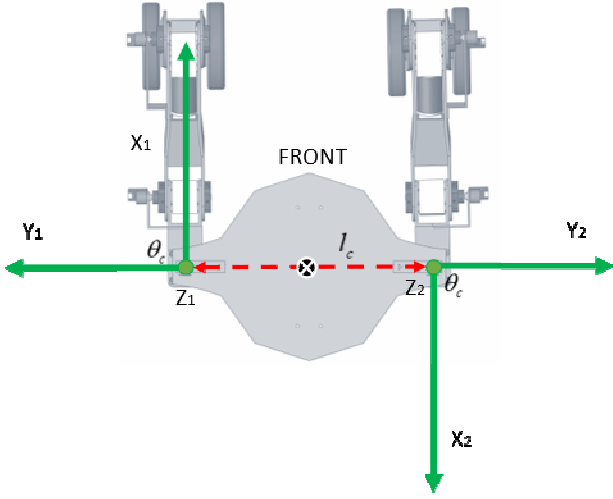


Fig. 2. BWLS's leg initial based on CCS.

B. Inverse and Forward Kinematic Models

In terms of kinematic chain and orientation, each leg of the BWLS operates as a direct-drive arthropod with a four degree of freedom (DoF) configuration. Position sensors are strategically placed directly at each joint, denoted as q_k with $k = 1, 2, \dots, 4$. In the geometric perspectives of each leg of BWLS can be computed using inverse kinematics (IK) as angular input translation as can be expressed as follows;

$$q_1 = \tan^{-1}\left(\frac{x}{y}\right) \quad (3)$$

$$q_2 = \frac{\pi}{2} \tan^{-1}\left(\frac{z - l_4}{\sqrt{x^2 - y^2}}\right) - \tan^{-1}\left(\frac{l_3 \sin \theta_3}{l_2 - l_3 \sin \theta_3}\right) \quad (4)$$

$$q_3 = \cos^{-1}\left(\frac{x^2 + y^2 + (z - l_4)^2 - l_2^2 - l_3^2}{2l_2l_3}\right) \quad (5)$$

$$q_4 = \pi - \theta_2 - \theta_3 \quad (6)$$

The orientation of the BWLS's leg is defined within the Coxa coordinate system (CCS), as depicted in Fig.3, incorporating links (l_k), joints (q_k), and torques (τ_w, τ_q) to establish precise movement.

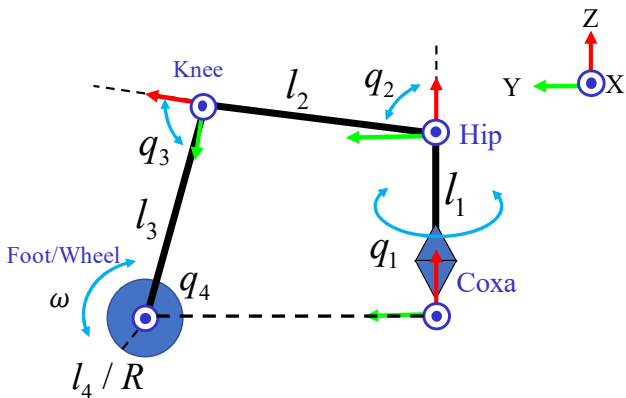


Fig. 3. BWLS's leg coordination and orientation that based on Coxa.

Cartesian coordinates, represented as $\Gamma: \{x, y, z\}$, define both the origin and end-effector of the leg. The Denavit-Hartenberg (D-H) conversion is utilized to formulate equations for forward kinematics (FK) that can be expressed as follows;

$$x = \alpha_1(l_1 + l_2\alpha_2 + l_3\alpha_{23} + l_4\alpha_{234}) \quad (7)$$

$$y = \beta_1(l_1 + l_2\alpha_2 + l_3\alpha_{23} + l_4\alpha_{234}) \quad (8)$$

$$z = l_2\beta_2 + l_3\beta_{23} + l_4\beta_{234} \quad (9)$$

where $\alpha_{ijk} = \sin(q_i + q_j + q_k)$ and $\beta_{ijk} = \cos(q_i + q_j + q_k)$ with $i = j = k = 1, 2, \dots, 4$. These kinematic expressions facilitate the translation of motion from Cartesian coordinates to q_k and vice versa, governing the movement of the BWLS's leg end-effector with meticulous precision. For the rolling motion, the linear velocity of the wheel (v_w) and angular speed (ω) are considered.

C. Trajectory Input Motion

For the initial assessment of this model, a straightforward standing trajectory input is devised, with a flat surface serving as the robot's floor. The axial motion during the standing phase for each leg can be defined as transitioning from equations (10) to (12), with the standing-up period (T) encompassing movement from the initial Cartesian position of each leg (Γ_0) to the designated setup height (h_0) as follows;

$$x = x_0 \quad (10)$$

$$y = y_0 \quad (11)$$

$$z = z_0 - h_0 \left(\frac{t}{T}\right) \quad (12)$$

Conversely, for the wheel rolling input the constant rotational angular speed by time is given to the q_4 for each leg, thus, it can be expressed as in (13) as follows;

$$\omega = \dot{q}_4 \quad (13)$$

The HCF for the BWLS is shown in Fig. 4. Here, the IK is used as axial trajectory input motion (Γ_{in}) to the joint angular motion for each revolute joint (q_{kin}) of the BWLS SM model, and FK is used as angular feedback translation (q_{kout}) to the end-effector axial (Γ_{out}) translation. A Proportional, Integral, and Derivative (PID) controller is employed for each joint position control, and simple trajectory inputs for initialization and standing up are designed for this study.

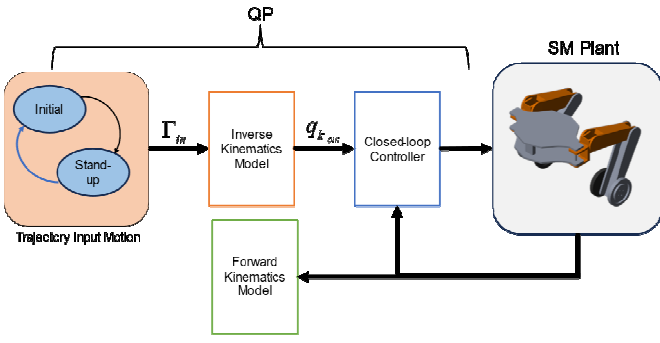


Fig. 4. Hierarchical Control Framework for BWLS robot

IV. RESULT AND DISCUSSION

A. Verification of Kinematics Models and Initial Input Determination

The initial verification process focused on validating the kinematic model's reliability for integration within the robot control frameworks.

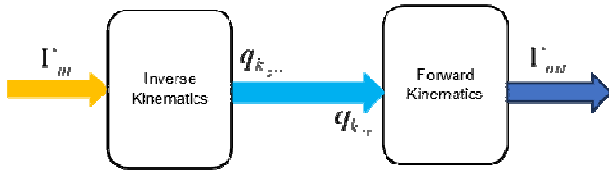


Fig. 5. Kinematic models verification flows.

This preliminary assessment involved a straightforward procedure: constant Cartesian values were assigned to the inverse kinematics (IK), and the resulting angular outputs were linked to the corresponding forward kinematics (FK) inputs, as illustrated in Fig. 3. Random Γ_{in} were subsequently introduced to the IK model, generating output values denoted as $q_{k,out}$ which were then matched with the FK inputs, $q_{k,in}$, ensuring $q_{k,out} = q_{k,in}$. The process culminated in an evaluation, as depicted in Fig. 4, where it was observed that $\Gamma_{in} \approx \Gamma_{out}$, indicating the congruence between input and output.

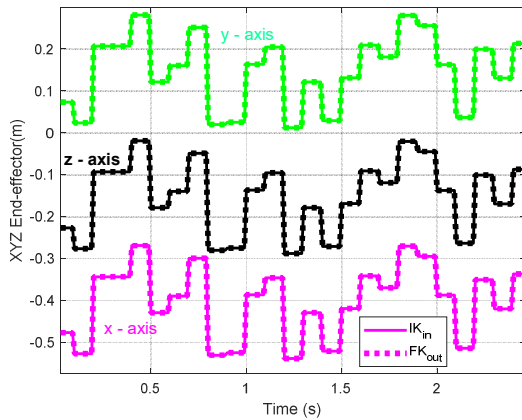


Fig. 6. Sample of $\Gamma_{in} \approx \Gamma_{out}$ (dotted line) for kinematic models verification.

This outcome serves to affirm the stability and efficacy of the kinematic models for integration within the robot's control frameworks. Noted that $\Gamma_{in} : \{x_i, y_i, z_i\}$ and $\Gamma_{out} : \{x_o, y_o, z_o\}$.

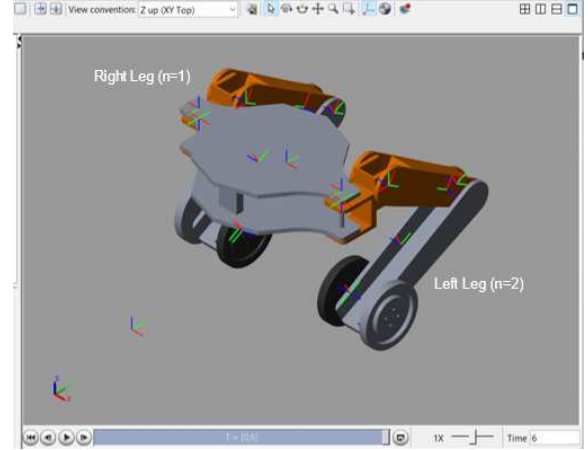


Fig. 7. SM for BWLS robot

For the next step of modeling, the initial position of the BWLS robot leg is determined manually through the SM model as shown in Fig. 6. This SM model has been translated from the conceptual model of BWLS from 3D solid drawings in Solidworks® according to actual structure in Fig.1. The initial position determination is done by using the slide bar input for each joint of the robot model and this process also revising the range of each joint in Table I. Table II shows the initial joint values, q_{kn_o} for each robot's joint according to the model in Fig. 6, the revised range of each joint angle and the initial Cartesian values, Γ_{n_o} , for each n -leg with $n = 1, 2$. Γ_{n_o} is determined by inserting the value of q_{kn_o} into (7),(8) and (9).

TABLE II. BWLS ROBOT'S INITIAL PARAMETERS AND REVISED JOINT RANGES

q_{kn_o} (Deg)	Range of angle (Deg) (Revised)	Γ_{n_o} (m)
$q_{11} = q_{12} = -180^0$	$-360^0 \leq q_{11} \leq -180^0$ (Leg 1) $-180^0 \leq q_{21} \leq 0^0$ (Leg 2)	0
$q_{21} = q_{22} = 120.2^0$	$80^0 \leq q_{21}, q_{22} \leq 140^0$	0.25
$q_{31} = 300.5$ $q_{32} = 59.28^0$	$220^0 \leq q_{31} \leq 320^0$ $42^0 \leq q_{32} \leq 138^0$	-0.6
$q_{41} = q_{42} = 60^0$	$10^0 \leq q_{41}, q_{42} \leq 90^0$ (walk mode)	

However, for the CCS deployment, only the z_{o_n} value is taken into consideration, while for other initial axes, (1) and (2) are used. Both individual legs used similar initial axis values as listed in Table II.

B. Basic Movement Test: Sprawling Coxa and Standing up trajectory

The SCS kinematic models, both FK and IK, are further verified by deployment on the SM model, as illustrated in Fig. 4. The initial testing involved injecting periodic waves, according to Coxa's angular range as listed in Table II, to verify the sprawling mechanism in the BWLS robot, as shown in Fig. 8. The results depicted in Fig. 9 demonstrate that the range of end-effector motion is similar for both legs, thus confirming the validity of the kinematic model.

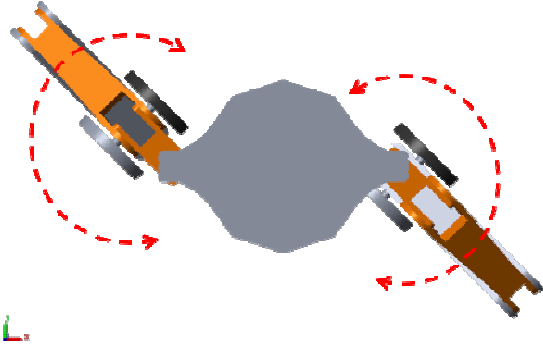


Fig. 8. Snapshot of SM simulation for sprawling motion test.

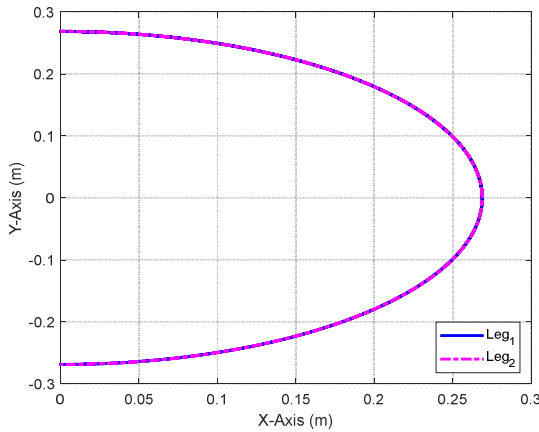


Fig. 9. Sample of X-Y end-effector (foot/wheel) motion rotating about Z-axis for both Leg 1 and Leg 2 of the robot.

Subsequently, additional validation was conducted by deploying the initial and stand-up trajectory discussed in Section IV(B). As shown in the SM snapshot in Fig. 10, the robot's legs move from the initial point as depicted in Fig. 10(a) to the set stand-up position shown in Fig. 10(b), with $h_0 = 0.025m$ for this simulation. This value adheres to the prescribed limitation for the BWLS, representing the stable length for its standing position.

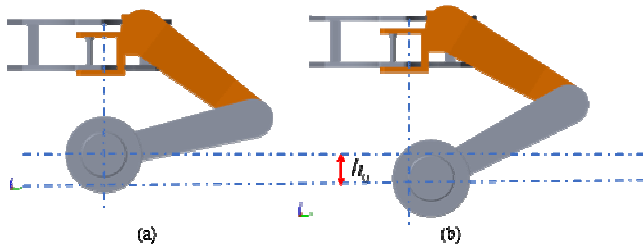


Fig. 10. Snapshot of SM simulation in side view; (a) Initial position, (b) Leg position after stand-up input.

The performance of this simulation can be depicted in the motion plots as shown in Fig. 11 and 12. From an angular motion perspective, the value decreased to about 120° , while another value increased to about 60° whenever Leg 2 started to move for the stand-up process, indicating some increases in z-axis motion as shown in Fig. 12.

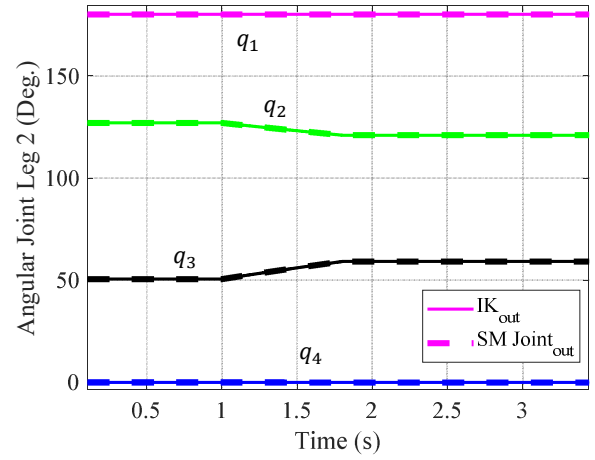


Fig. 11. Sample of angular joint motion (Leg 1) setpoint trajectory vs. measured joint (dotted line)

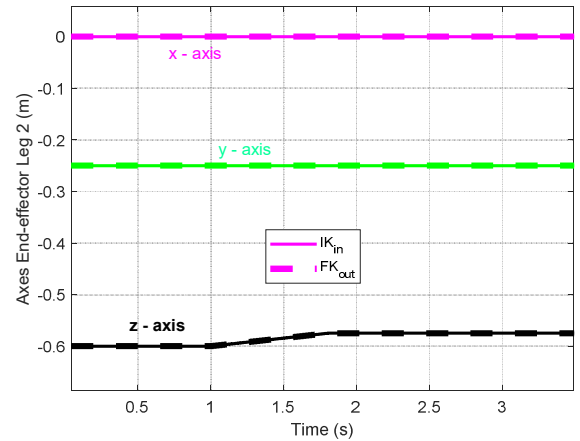


Fig. 12. Sample of end-effector axes positioning (Leg 1) setpoint trajectory vs. measured joint (dotted).

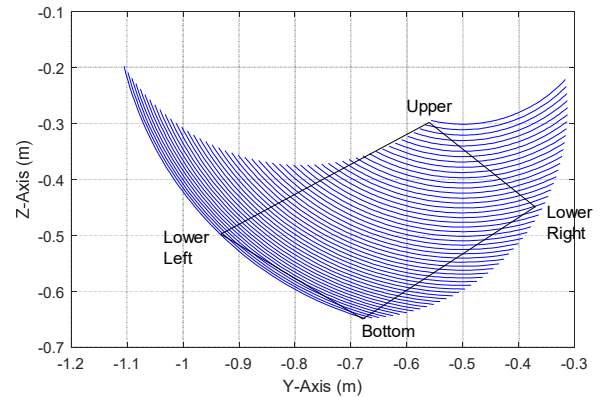


Fig. 13. Working space of BWLS leg with optimum foot/wheel position points

The working space for a leg of BWLS can be represented by the Y-Z space as shown in Fig.13. Here, the working space is measured from the sliding manipulability of the leg, where the initial values for the first three angles and the range for each of the first three angles are evaluated according to the FK. Four points can be identified as the vertices of the trapezium boundary for the stable attitude for the robot's locomotion. The setup shows that the z-axis motion remains within the stable boundary after the standing-up movement.

V. CONCLUSION

The modeling of the BWLS robot using the SM and proposed CCS kinematics model is presented. The reliability of these kinematic models has been verified through the transition to a SM model and the application of the proposed CCS. This establishment is achieved by developing a HCF to deploy the proposed CCS for the SM model of the BWLS, using simple trajectory inputs. The determination of the range of angles and initial positioning of the angular joints aids in identifying the stable boundary of the leg through FK calculation. Future tasks will focus on further refining the model by incorporating contact force formulations into the SM and developing additional control strategies to optimize the stability of the robot's locomotion and skiing movements.

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